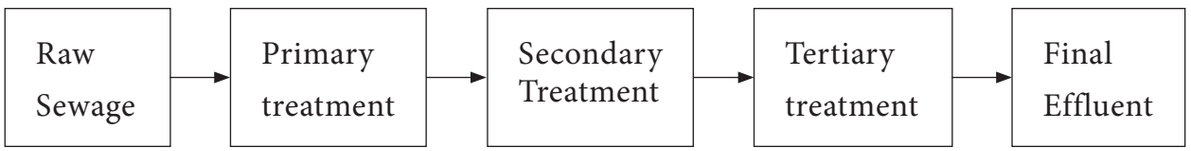
World's Largest Sewage Treatment Plant



Stickney Water Reclamation Plant located in Chicago is the largest sewage plant in the world. It has a daily capacity of 1.44 billion gallons. It's owned by Metropolitan Water Reclamation District (MWRD).

Flow diagram of sewage treatment system



5.6.1 Primary treatment

The main aim of this treatment is to remove floating and settleable solids from waste water. It includes physical operation such as screening and sedimentation.

a. Screens

In this method, sewage is passed through different sizes of screens. It is used to retain solid materials such as cloth, paper, wooden material, rubber, cork, etc.

b. Skimming tanks

Skimming tanks are used to remove oil and grease from the sewage. It is placed before the sedimentation tanks.

c. Grit chamber

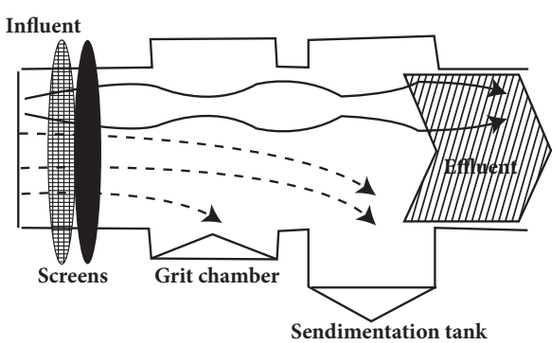
Grit chamber are used to remove inorganic particles such as sand, gravel, grit, egg shells, etc. They are located before pumping station and after the screen.

d. Sedimentation tank

Very fine suspended inorganic matters can be removed only in sedimentation tanks. When the velocity of water is reduced in sedimentation tanks the suspended particles settles down.

According to location, sedimentation tanks can be classified into two types

1. **Primary clarifiers**-They are located just after the grit chamber
2. **Secondary clarifiers**-They are located after secondary treatment units.



Primary Treatment

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